

Towards Robust Deepfake Detection Based on Heart Rate Analysis

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Abstract

Remote photoplethysmography (rPPG)-based deepfake detection analyzes a target's heart rate (HR) patterns in a video. However, existing studies on rPPG-based deepfake detection have primarily focused on detection algorithms that utilize extracted HR values, rather than on mechanisms for acquiring and analyzing informative HR patterns. To establish robust rPPG-based deepfake detection, it is required to clarify the impact of key factors that directly affect HR extraction and processing. We defined the key factors that directly affect the performance of rPPG-based deepfake detection, including the facial region, extraction interval, rPPG extraction method, and feature engineering method. Based on these defined key factors, we assessed the impact of each variable for the key factors on rPPG-based deepfake detection performance. Furthermore, we identified the optimal combination of variables, which maximizes the performance of rPPG-based deepfake detection. Through statistical validation of 164 real-world videos, we conducted experimental evaluations. The results revealed variables for each key factor that yielded significant differences in HR values between real and fake videos. Moreover, we identified the optimal combination of the variables, enabling a pronounced distinction between real and fake videos. This paper successfully confirms the impactful role of individual variables and their optimal combinations for each key factor in rPPG-based deepfake detection.

Keywords: Deepfake, deepfake detection, remote photoplethysmography, rPPG, face recognition

1. Introduction

Deepfakes refer to manipulated media that appear authentic by merging, replacing, or superimposing videos onto an existing video source [1]. The emergence of deepfakes has introduced new security threats such as the dissemination of fake news, manipulation of public opinion, and exploitation of vulnerabilities in face recognition systems [2, 3]. As deepfake generation techniques advance, it has become increasingly challenging to verify the authenticity of videos.

Recently, deepfake detection methods have begun to leverage biological signals extracted from videos, such as heart rate (HR), head movement, and eye blinking [4, 5]. In particular, remote photoplethysmography (rPPG) has emerged as a prominent method for detecting deepfakes [6]. rPPG is a non-contact video analysis technique that measures an individual's HR by analyzing pixel intensity changes (i.e., RGB values) from their skin, allowing for the analysis of changes in blood volume [7]. rPPG-based deepfake detection methods extract HR patterns from a video, utilize them to train a classifier, and identify abnormal HR patterns [8, 9]. The mechanism of processing the extracted HR has a significant impact on the performance of rPPG-based deepfake detection. However, existing approaches have primarily focused on configuring solutions using the extracted HR data, rather than on extracting and analyzing informative HR data. In particular, the key factors that directly affect HR extraction and processing, as well as their impacts, have not been explicitly identified.

For instance, the type of rPPG extraction method can affect the measurement of HR patterns, which, in turn, can impact the performance of rPPG-based deepfake detection. Although image quality factors such as brightness, contrast, and resolution may also affect, it is uncommon to manipulate them while generating a deepfake. Therefore, to design a robust rPPG-based deepfake detection method, it is required to define the key factors that directly affect the extraction and processing of HR and to measure their impacts on rPPG-based deepfake detection. While image quality factors can presumably have an impact, this study does not take them into account as it is inherently challenging to manipulate them arbitrarily during deepfake generation.

In this paper, we identified four key factors that influence rPPG-based deepfake detection as follows: (1) The facial region for measuring HR (=facial region), (2) The time interval for extracting HR (=extraction interval), (3) The rPPG method for extracting HR (=rPPG extraction method), and (4) The method for processing extracted HR into features (=feature engineering method). We assessed the impact of each variable for the key factors on rPPG-based deepfake detection by measuring the difference between real and fake videos through statistical validation. We performed experimental evaluations with a total of 164 videos (82 real videos and 82 fake videos) selected from the FaceForensics++ dataset (FF++) [10]. Furthermore, we analyzed a set of key factor combinations to identify the optimal combination that maximizes the performance of rPPG-based deepfake detection.

This paper presents four pivotal contributions: (1) We defined the key factors influencing rPPG-based deepfake detection, thereby providing a systematic foundation for the field. (2) We conducted a thorough analysis of how each key factor impacts rPPG-based deepfake detection. (3) We assessed combinations of key factors, which maximize rPPG-based deepfake detection performance. (4) We performed rigorous evaluations involving 164 real-world videos, which not only validate the study but also underscore its practical relevance.

This paper is organized as follows. Section 2 discusses related work. Section 3 describes our approach, and Section 4 illustrates the results of our experimental analysis. Section 5 presents a discussion, and Section 6 concludes this study.

2. Related Work

2.1 Deepfake Generation

Deepfake generation methods are classified into two categories according to the deepfake generation principles: Variable Auto Encoder (VAE) and Generative Adversarial Network (GAN). VAE-based generation methods generate deepfakes by extracting features with an encoder and reconstructing faces using the features with a decoder. VAE-based generation methods include FakeApp and DeepFakes [11, 12]. FakeApp automatically extracts faces from the source and target videos and identifies similarities between them [13], while DeepFakes extracts features from the source video using an encoder and reconstructs deepfakes using a decoder trained on the target video [14]. In contrast, GAN-based methods generate realistic deepfakes by repeatedly generating fake videos with a generator and determining their authenticity with a discriminator [15]. GAN-based generation methods include Face Swapping GAN (FSGAN) and StyleGAN [16, 17]. FSGAN generates deepfakes by making the source video's mask similar to the target video's mask, and inpainting to fill the margins [16]. StyleGAN generates deepfakes with desired visual attributes by applying style information to each layer of the generator [17]. GAN-generated deepfakes present challenges for both face recognition systems and existing detection methods [18]. FaceShifter generates deepfakes by synthesizing the face from the source video onto the target video, swapping the identities between the source and target videos using a GAN-based adaptive embedding integration network [19].

2.2 Deepfake Detection

Depending on the generation methods, deepfakes can have different types of artifacts that serve as valuable information for deepfake detection. The artifacts used in deepfake detection methods can be classified into two categories: visual artifacts and temporal artifacts.

- **Visual artifacts.** Fake videos are synthesized by warping the face to maintain the configuration of source videos, which can result in visual artifacts due to resolution inconsistencies between the face and surrounding contexts. Furthermore, if global consistency is lacking or if the lighting or the face angle in the video is incorrectly estimated during the synthesis process, visual artifacts such as missing reflections and lack of details in facial areas can be generated [20]. Li et al. proposed a method to detect deepfakes by analyzing visual artifacts that exhibit distinct resolutions around the facial boundary [21]. However, because the visual artifacts produced by fake videos are primarily determined by the deepfake generation methods, the visual artifacts-based deepfake detection methods are highly dependent on deepfake generation methods [22].
- **Temporal artifacts.** Videos consist of a series of frames that exhibit temporal consistency between them. However, deepfake generation methods do not primarily consider the temporal consistency between frames during the synthesis process. This can result in temporal artifacts such as flickering in certain areas or inconsistent facial expressions in the video [23]. Tursman et al. proposed a method to detect deepfakes by analyzing changes in geometry around the speaker's mouth [24]. However, as recent deepfake generation methods begin to consider temporal consistency [25–27], it is becoming more challenging to distinguish new types of fake videos from existing detection methods.

Recently, deepfake detection methods using biological signals (e.g., head positions and eye

blinking) have been proposed. These methods take into account that deepfakes are generated without reflecting human traits or physiological responses and detect deepfakes by identifying visual artifacts (e.g., incorrect head poses), and temporal artifacts (e.g., lack of eye blinking). Yang et al. proposed a method to detect fake videos using head pose information estimated from 2D facial landmarks. This method is based on the fact that facial landmarks (e.g., the positions of the eyes and mouth tips) are not maintained during the synthesis process [5]. Li et al. proposed a method to detect fake videos by analyzing the lack of eye-blinking information. They argued that eye blinking rarely appears in a real image used to generate fake videos [4]. Jung et al. proposed a method to detect fake videos by analyzing changes in a person's eye blinking pattern in the video, including factors such as blink duration and frequency [28].

2.3 Deepfake Detection using rPPG

The heart changes the blood flow inside the body every time it beats, and the HR is synchronized accordingly. Therefore, changes in blood flow can be used to measure HRs [29]. The pulsating blood that spreads through the human cardiovascular system changes the blood volume of the skin tissue, resulting in a subtle color change on the skin surface. RGB cameras can detect these subtle color changes that occur on the skin surface every time the heart beats. The principle of rPPG is to detect these color changes in the skin surface as RGB values and calculate the RGB values as HRs [30]. With the development of rPPG technology, it is now feasible to obtain high-quality signals by accurately extracting HRs. Notably, the extracted HRs may exhibit distinguishable patterns between real and fake videos, serving as key features for deepfake detection. Subsequently, several rPPG-based deepfake detection methods that leverage these distinctive features have been proposed. Boccignone et al. introduced the pyVHR framework, which supports various rPPG extraction methods [31]. They compared the extracted HR values with the actual HR values using a pulse rate detection dataset [32]. *CHROM* exhibited the lowest mean absolute error (MAE) value of 3.27, while others—*LGI*, *PBV*, *PCA*, *GREEN*, and *ICA*—showed MAE values of 7.29, 4.14, 3.87, 10.19, and 9.38, respectively. As a result, studies that apply rPPG to deepfake detection have been proposed. Ciftci et al. proposed a machine learning-based deepfake detection model trained with feature vectors and photoplethysmography (PPG) maps, respectively. The PPG maps were generated by transforming biological signals extracted from a specific region of interest (ROI) [9]. Qi et al. proposed a machine learning-based deepfake detection model trained with features obtained by processing heart rhythms extracted from both real and fake videos [8]. Wu et al. proposed a two-stage network that applies both a local attention module and a temporal transformer to detect deepfakes by identifying spatial and temporal inconsistencies in PPG maps of videos [33]. Previous rPPG-based deepfake detection methods have focused on extracting and processing HR information in specific ways and generating machine learning-based detection models trained with processed information. However, the main factors used in HR extraction and processing are not clearly defined, and the impact of each factor on deepfake detection performance is not clarified.

3. Our Approach

3.1 Overview

The study aimed to identify key factors for rPPG-based deepfake detection. To achieve this goal, experimental analyses were conducted to investigate the impact of these key factors on the distinction between real and fake videos and determine their optimal combinations for

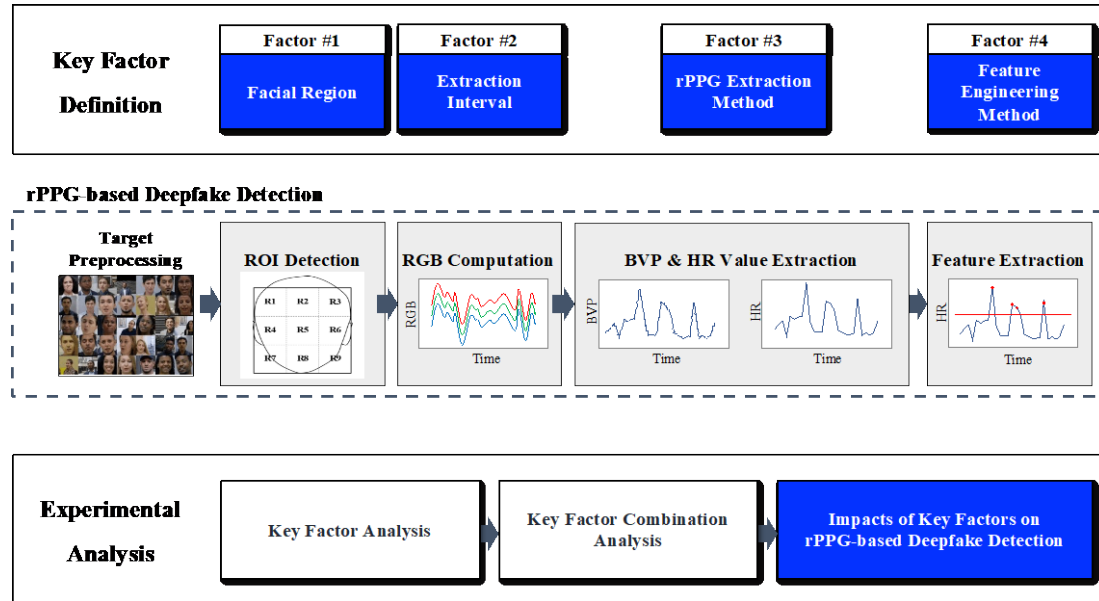


Fig. 1. Overview of our approach

rPPG-based deepfake detection. Fig. 1 provides an overview of our approach, which consists of two phases: key factor definition and experimental analysis.

3.1.1 Key Factor Definition

To the best of our knowledge, this study is the first to define and analyze key factors for rPPG-based deepfake detection. We systematically reviewed the existing literature published in reputable journals and conferences to identify the key factors involved in rPPG-based deepfake detection. Note that we carefully followed the general guidelines for a systematic literature review process [34]. The first step involved conducting a keyword-based literature search that targeted two domains: deepfake generation and deepfake detection using rPPG. The search engines used for this purpose included IEEE Explore, ACM Digital Library, Springer Link, and Google Scholar. The keywords for the search query were formed by combining the domain keywords and their synonyms (e.g., rPPG and remote photoplethysmography), and the scope of the search was limited to literature published after 2015. Consequently, a total of 973 publications were initially collected, with 657 related to deepfake generation and 316 related to deepfake detection using rPPG technology. Afterward, we conducted a brief literature review by examining whether the title and abstract of each literature contained words related to the keywords (e.g., for “deepfake generation”: “generate”, “obstruct”, “faceswap”, and “impersonation”). Finally, 55 literatures were selected.

We conducted a detailed literature review focusing on the mechanism of signal extraction using rPPG and its processing for deepfake detection. Based on the review, we identified the fundamental steps involved in rPPG-based deepfake detection, which include: (a) Detecting a region of interest (ROI) from video frames using a face detection framework. In general, the ROI is set to either the entire face or specific facial regions such as the forehead, cheeks, and chin; (b) Extracting pixel values from the ROI’s RGB channels and measuring the average color intensity over a predefined interval; (c) Estimating RGB values as blood volume pulses (BVP) using an rPPG extraction method, where BVP is a value associated with blood volume

Table 1. Key factors for rPPG-based deepfake detection

No.	Key Factor	Definition	Variable
#1	Facial Region	A facial area from which RGB pixel values are extracted.	<i>R1, R2, R3, R4, R5, R6, R7, R8, R9, Avg</i>
#2	Extraction Interval	A time interval that divides a video into frames.	<i>1 second, 0.5 seconds, 0.1 seconds</i>
#3	rPPG Extraction Method	A method that transforms RGB pixel values into BVP.	<i>CHROM, GREEN, ICA, LGI, PBV, PCA</i>
#4	Feature Engineering Method	A method that extracts discriminative features for deepfake detection from HR values.	<i>feat#1, feat#2, feat#3</i>

changes in blood vessels. A short-time Fourier transforms (STFT) algorithm is then applied to transform the BVP signal into a power spectrum density (PSD), and HR values are estimated by selecting the peak of the PSD in the frequency domain; and (d) Finally, by processing the calculated HR values, an rPPG-based deepfake detection method extracts relevant features.

Each step requires specific configuration settings that directly influence the extracted HR values, consequently impacting the overall performance of rPPG-based deepfake detection. We assessed these configuration settings and identified four key factors, each corresponding to one of these steps, as shown in **Table 1**: (a) A “facial region” (Factor#1) refers to a specific area from which RGB pixel values are extracted from each frame in a video. As RGB pixel values vary depending on facial regions, the HR values can differ for each region; (b) An “extraction interval” (Factor#2) is the time interval during which pixel values are extracted from a video. As RGB pixel values vary depending on extraction intervals, the HR values can differ for each interval; (c) An “rPPG extraction method” (Factor#3) refers to a mechanism that converts RGB pixel values into BVP. As each method operates using different calculation mechanisms, BVP can be extracted differently, resulting in different HR values; and (d) A “feature engineering method” (Factor#4) refers to a mechanism that extracts discriminative features for deepfake detection from HR values. The distinction between real and fake videos can differ depending on the mechanism used to process the HR value into features.

3.1.2 Experimental Analysis

Based on the defined key factors, we performed the fundamental steps for rPPG-based deepfake detection as shown in **Fig. 1**. The details of each step are as follows. (1) Target Preprocessing: We assessed datasets for deepfake detection and selected FF++, the most popular dataset containing both source and target videos [10]; (2) ROI Detection: For each data, the facial boundary was defined as follows: top (hair line), bottom (chin contour), and left-right (face contour). A total of ten computer science researchers, including the authors, manually labeled and double-checked the facial boundary of each data. The detected facial area is then segmented into 9 regions, each of which could be an ROI for HR extraction; (3) RGB Computation: We measured the average color intensity from each ROI and calculated RGB values based on predefined extraction intervals; (4) BVP Computation: With the extracted RGB values, we estimated BVP, which was then transformed to HR values using the

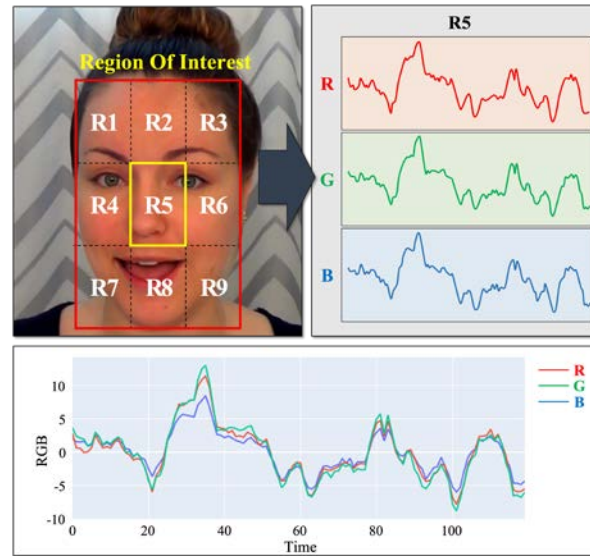


Fig. 2. Examples of extraction RGB values from facial region R5

STFT; and (5) Feature Extraction: We converted the HR values into feature values using the feature engineering methods defined in this paper. Throughout these steps, we analyzed the experimental findings using statistical validation. Specifically, for each key factor, we analyzed the difference in HR values between real and fake videos using an independent sample t-test [35, 36] based on the following statistical metrics: The *t-value* represents the difference between the means of the two groups, and the degrees of freedom (df) indicate the variability within each group. The *p* indicates the probability of the observed values occurring by chance, typically with a value below 0.05 implying a correlation between the two groups. We then analyzed a set of key factor combinations and identified the optimal one that optimizes the effectiveness of rPPG-based deepfake detection.

3.2 Factor Definition

3.2.1 Facial Region

As the positions of blood vessels, blood volume, and skin color vary across different facial regions, the rPPG extraction results—specifically, HR values—calculated from different facial regions (i.e., forehead, cheek, and chin) may vary [31, 37–39]. As rPPG primarily extracts information from the exposed skin area of the face, we did not consider facial features such as eyes, mouth, or eyebrows as criteria when dividing the facial area. Instead, we only considered the minimum required extraction area. As shown in Fig. 2, in this study, we divided the facial area equally into 3×3 regions, comprising a total of 9 regions. For instance, R2 represents the forehead region, whereas R8 represents the chin region. In this study, we selected ten variables for the facial regions as follows: R1–R9, which equally divide a facial area, and their average (=Avg). For each facial region, we analyzed the difference between real and fake videos based on the extracted HR values.

3.2.2 Extraction Interval

Depending on the time intervals for extraction, rPPG extraction results may differ. For example, CHROM [40], a popular rPPG extraction method, can produce its most accurate

Table 2. Description of rPPG extraction methods

Method	Description
<i>CHROM</i> [40]	<i>CHROM</i> is a chrominance-based rPPG method for minimizing image distortion. <i>CHROM</i> calculates BVP as a linear combination of three-color channels for pulses under the assumption that skin tone is standardized.
<i>GREEN</i> [7]	<i>GREEN</i> is an rPPG method that calculates BVP by averaging the values of the green (G)-channel. Compared to other channels, the G-channel can more accurately represent the characteristics of HR because hemoglobin in blood absorbs green light better than red and blue light.
<i>ICA</i> [41]	<i>ICA</i> is an rPPG method for analyzing biomedical signals to extract rPPG and remove noise when measuring conduction or electroencephalography. <i>ICA</i> estimates BVP by removing artifacts caused by motion or ambient light conditions and reconstructing the source signals.
<i>LGI</i> [45]	<i>LGI</i> is an rPPG method that is robust to nuisance factors (i.e., head motion, talking, and facial expressions). <i>LGI</i> estimates HR using features that are invariant to nuisance factors while preserving target information.
<i>PBV</i> [46]	<i>PBV</i> is an rPPG method that is robust to the target's motions by minimizing the assumptions about the periodicity of the pulse signal and distortion. <i>PBV</i> estimates HR using the characteristics that arterial blood and bloodless skin have different absorption spectra.
<i>PCA</i> [47]	<i>PCA</i> is an rPPG method that is based on focusing on similarities and differences in data. <i>PCA</i> estimates heart rate-related changeable components using RGB values filtered by the FIR bandpass filter.

result with an interval of 1.6 seconds, while *ICA* performs better at longer intervals of 25 seconds. If the extraction interval is set too long, which directly affects the extraction latency, the results of HR extraction can be inaccurate [40]. In this study, considering the typical length of fake videos, we selected three variables for the extraction interval as follows: 1 second, 0.5 seconds, and 0.1 seconds. For each extraction interval, we analyzed the difference between real and fake videos based on the extracted HR values.

3.2.3 rPPG Extraction Method

The general rPPG extraction process for obtaining HR values is as follows [41–44]: (1) Raw RGB values are calculated after extracting an ROI from the detected facial area; (2) To remove noise, the raw RGB values are subjected to signal preprocessing (e.g., detrending and filtering); (3) From the signals with noise removed, BVP is estimated using rPPG methods (e.g., *CHROM* [40] and *ICA* [41]); (4) PSD is calculated from the BVP using the STFT; and (5) Finally, HR values are estimated by selecting the peaks from the PSD in the frequency domain. As demonstrated in Table 2, several rPPG extraction methods have been proposed [7, 40, 41, 45–47], each of which is built on a distinct algorithm for obtaining BVP from RGB values. For example, *CHROM* and *PBV* extract higher quality signals for HR values than *ICA* and *PCA* from a video of a subject exercising for fitness [48]. In this study, we selected six variables for the rPPG extraction method as follows: *CHROM*, *GREEN*, *ICA*, *LGI*, *PBV*, and *PCA*. For each rPPG extraction method, we analyzed the difference between real and fake videos based on the extracted HR values.

3.2.4 Feature Engineering Method

Previous studies have shown that HR patterns in fake videos are distinguishable from those in real videos due to their inconsistent HR information [8, 9]. Since existing rPPG-based deepfake detection methods analyze particular HR patterns as features, their detection performance may vary depending on the way their feature type is defined. Depending on the target approach, different types of metrics have been proposed for analyzing the human autonomic nervous system using HR information [49]. Among the standard metrics for HR variability analysis, we adopted three representative metrics that measure time-domain features: SDNN, RMSSD, and SDDSD [50, 51]. The standard deviation of NN intervals (SDNN) reflects both long-term factors (e.g., baroreceptor reflexes) and short-term factors (e.g., respiration), which determine HR variability. Meanwhile, the root mean square of successive RR interval differences (RMSSD) and the standard deviation of successive RR interval differences (SDDSD) mainly reflect short-term HR variability (e.g., high-frequency variation). Based on the metrics, each feature engineering method in this study was designed to capture the overall variability of HR values or variability between successive HR values. The details of each method are as follows.

feat#1 analyzes the standard deviation of all HR values, which is inspired by SDNN (see (1)) [52]. The differences between the HR value at time t (h_t) and the mean HR value ($=m_t$) are summed for the entire video length and divided by l (where t is a temporal point defined by the extraction interval and l is the total length of the video). Then, the square root of this sum is calculated to measure the standard deviation of all HR values.

$$feat\#1: \sqrt{\frac{\sum_{t=0}^l (h_t - m_t)^2}{l}} \quad (1)$$

feat#2 analyzes the root mean square of differences between adjacent HR values, which is inspired by RMSSD (see (2)) [52]. The differences between the HR value at time $t + 1$ ($=h_{t+1}$) and the HR value at time t ($=h_t$) are summed for the entire video length and divided by $l - 1$ (where l is the total length of the video). Then, the square root of this sum is calculated to measure the root mean square of differences between adjacent HR values.

$$feat\#2: \sqrt{\frac{\sum_{t=0}^{l-1} (h_{t+1} - h_t)^2}{l-1}} \quad (2)$$

feat#3 analyzes the standard deviation of differences between adjacent HR values, which is inspired by SDDSD (see (3)) [52]. The differences between the HR value at time $t + 1$ ($=h_{t+1}$), the HR value at time t ($=h_t$), and the mean of differences ($=m_d$) are summed for the entire video length and divided by $l - 1$ (where l is the total length of the video). Then, the square root of this sum is calculated to measure the standard deviation of differences between adjacent HR values.

$$feat\#3: \sqrt{\frac{\sum_{t=0}^{l-1} ((h_{t+1} - h_t) - m_d)^2}{l-1}} \quad (3)$$

For each feature engineering method (i.e., *feat#1*, *feat#2*, and *feat#3*), we analyzed the difference between real and fake videos based on the extracted HR values.

Table 3. Control variables for each key factor

Key Factor	Control Variables			
	Factor#1	Factor#2	Factor#3	Factor#4
Factor#1	-	<i>0.1 seconds</i>	<i>GREEN</i>	<i>feat#1</i>
Factor#2	<i>R5</i>	-	<i>GREEN</i>	<i>feat#1</i>
Factor#3	<i>R5</i>	<i>0.1 seconds</i>	-	<i>feat#1</i>
Factor#4	<i>R5</i>	<i>0.1 seconds</i>	<i>GREEN</i>	-

3.3 Dataset

This study used FF++, which is one of the most widely used open deepfake datasets for deepfake detection studies [10]. The FF++ comprises videos compressed at different compression ratios (i.e., raw, c23, and c40). Considering that noise introduced by video compression may affect the quality of the HR signal, we selected the uncompressed videos (i.e., raw) as our target dataset [31, 53, 54]. The FF++ raw dataset includes a total of 8,068 fake videos generated using various deepfake generation methods, along with 1,363 real videos either recorded directly or collected from YouTube, providing diverse range of real-world conditions [10]. To ensure consistency in the source videos, we selected fake videos generated from the same source videos using five different deepfake generation methods (i.e., Deepfakes, Face2Face, FaceShifter, FaceSwap, and NeuralTextures). Given that the facial region serves as the ROI in rPPG, accurate face detection is crucial. Hence, we applied the following criteria to select videos from the FF++ raw dataset: (a) Videos featuring individuals facing forward; (b) Videos without partial occlusion of faces, such as facial hair, glasses, or accessories; and (c) Videos with only one visible person at any given moment. Note that, considering that each fake video was generated based on two source videos, we only selected fake videos whose source videos both satisfy the criteria. As a result, we constructed a dataset consisting of 164 videos, 82 real videos, and 82 fake videos. To minimize potential bias, we ensured diversity in both subjects and scenarios when composing the dataset. The videos in the dataset encompass a variety of races, genders, age groups, and hair styles, as well as backgrounds, covering diverse lighting conditions and facial expressions. The details of the constructed dataset are publicly available at <https://github.com/HG-AISLAB/Heart-Rate-Analysis-for-rPPG-based-Deepfake-Detection>.

4. Experimental Results

In section 4.1, this paper analyzed the experimental results that demonstrated the impact of each variable for the key factors on rPPG-based deepfake detection. We defined the control variables based on the results of prior studies, specifically the performance comparison of variables in terms of HR extraction accuracy [48], and our preliminary study which identified significant variables ($p < 0.05$) through t-tests conducted on a total of 540 variable combinations. Consequently, as shown in Table 3, the control variables for each key factor were defined as follows: facial region ($=R5$); extraction interval ($=0.1$ seconds); rPPG extraction method ($=GREEN$); and feature engineering method ($=feat#1$). In section 4.2, this paper assessed a set of key factor combinations and determined the one that maximizes detection performance. In

our evaluations, rPPG extractions were performed using pyVHR 0.0.4 [31]. pyVHR is an open-source Python framework for rPPG extraction, which supports various rPPG extraction methods (i.e., *ICA*, *PCA*, *GREEN*, *CHROM*, *LGI*, and *PBV*). Operating as an end-to-end pipeline, pyVHR estimates the BPM of the facial region in the video through spectral analysis of the BVP signal. The BVP signal is derived from the pulse signal in the RGB space, extracted from the facial region in the video using the user-selected rPPG extraction method. All evaluations were conducted using a single GPU (GeForce RTX 3060) in a Python 3.6.9 environment.

4.1 Key Factor Analysis

In this section, this paper analyzed the experimental results and verified the impact of four key factors for rPPG-based deepfake detection.

4.1.1 Facial Region

We evaluated the differences between real and fake videos in terms of the extracted HR for each facial region (i.e., *R1–R9* and *Avg*). In the independent sample t-test ($p < 0.05$), *R5* offered

Table 4. The detailed results of the t-test on HR values between real and fake videos at different facial regions

Facial Region	<i>t-value</i>	<i>df</i> [†]	<i>p</i>
<i>R1</i>	0.901	161.846	0.369
<i>R2</i>	0.096	158.932	0.924
<i>R3</i>	0.300	161.773	0.765
<i>R4</i>	3.804	130.701	2.E-04*
<i>R5</i>	5.216	125.958	5.53E-07*
<i>R6</i>	2.191	161.720	0.030*
<i>R7</i>	2.372	154.246	0.019*
<i>R8</i>	0.614	151.643	0.540
<i>R9</i>	2.314	142.229	0.022*
<i>Avg</i>	2.781	135.362	0.006*

* $p < 0.05$, *df*[†]: degree of freedom

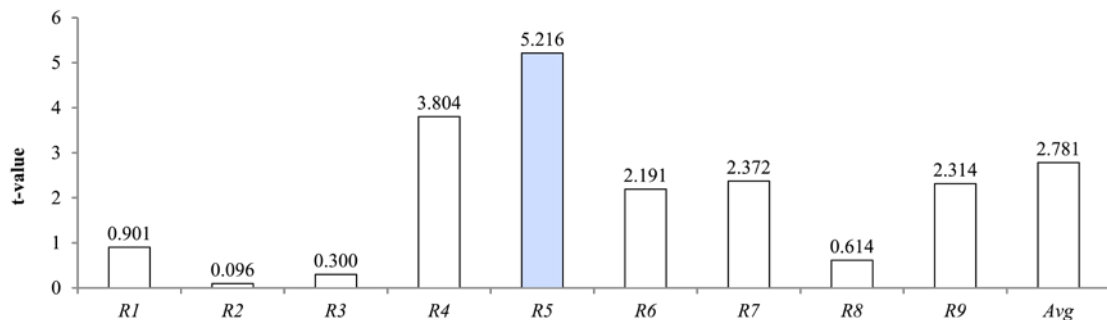


Fig. 3. The t-test results on HR values between real and fake videos at different facial regions

Table 5. The detailed results of the t-test on HR values between real and fake videos at different extraction intervals

Extraction Interval	<i>t</i> -value	<i>df</i> [†]	<i>p</i>
1 second	4.771	130.934	4.06E-06*
0.5 seconds	5.006	127.617	1.44E-06*
0.1 seconds	5.216	125.958	5.53E-07*

* $p < 0.05$, *df*[†]: degree of freedom

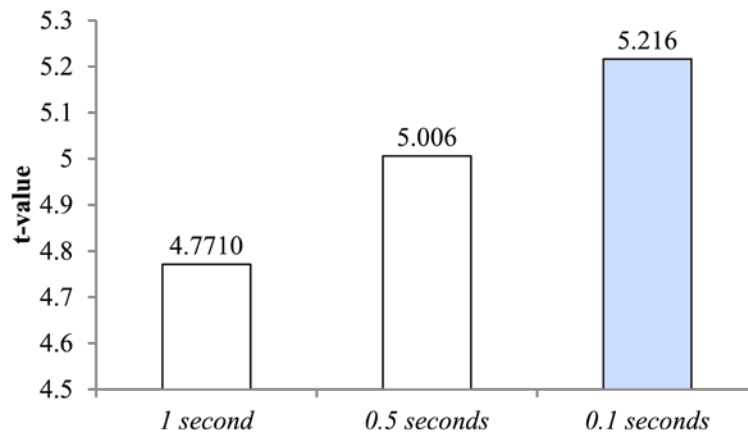


Fig. 4. The t-test results on HR values between real and fake videos at different extraction intervals

the largest *t*-value of 5.216, as shown in Table 4 and Fig. 3. These results demonstrate that, in rPPG-based deepfake detection, analysis at *R5* can more clearly distinguish between real and fake videos compared to analysis at other regions.

The experimental results indicate that *R5* had the largest *t*-value (=5.216) because it can contain shading artifacts generated around the nose. Among the various artifacts that can occur in fake videos, shading artifacts (e.g., unnatural shadows) are primarily generated around the nose [55]. These shading artifacts can cause abnormal patterns in HR extracted from fake videos, consequently facilitating the distinction between real and fake videos. As a result, our experiment demonstrates that the difference between real and fake videos is most apparent when using *R5* in rPPG-based deepfake detection, as the shading artifacts in *R5* can induce abnormal HR patterns in fake videos.

4.1.2 Extraction Interval

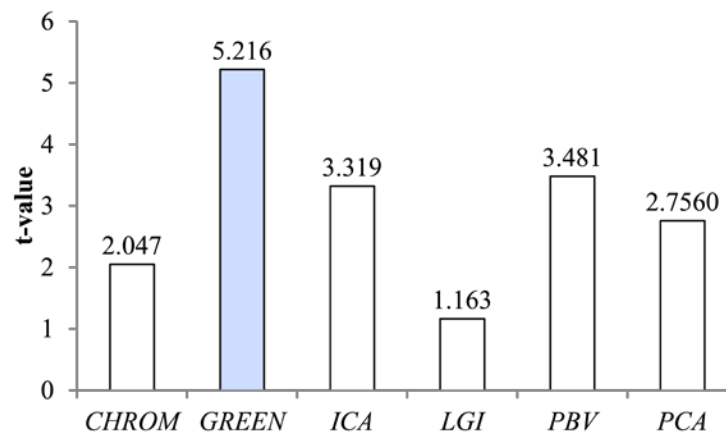
We evaluated the differences between real and fake videos in terms of the extracted HR for each extraction interval (i.e., 1 second, 0.5 seconds, and 0.1 seconds). In the independent sample t-test ($p < 0.05$), 0.1 seconds offered the largest *t*-value of 5.216, as shown in Table 5 and Fig. 4. These results demonstrate that, in rPPG-based deepfake detection, analysis using a 0.1 seconds interval can more clearly distinguish between real and fake videos compared to analysis using other intervals.

The experimental results indicate that 0.1 seconds had the largest *t*-value (=5.216) because it can analyze more frames for HR extraction. In rPPG-based HR extraction, a shorter extraction interval allows for the analysis of a greater number of frames, enabling more

Table 6. The detailed results of the t-test on HR values between real and fake videos using different rPPG extraction methods

rPPG Extraction Method	<i>t</i> -value	<i>df</i> [†]	<i>p</i>
<i>CHROM</i>	2.047	160.921	0.042*
<i>GREEN</i>	5.216	125.958	5.53E-07*
<i>ICA</i>	3.319	128.417	0.001*
<i>LGI</i>	1.163	159.056	0.246
<i>PBV</i>	3.481	115.319	6.E-04*
<i>PCA</i>	2.756	152.104	7.E-03*

* $p < 0.05$, $df^{\dagger}$: degree of freedom

**Fig. 5.** The t-test results on HR values between real and fake videos using different rPPG extraction methods

accurate detection of subtle temporal relationships and facilitating more precise extraction [56]. In particular, fake videos may exhibit temporal inconsistencies between frames as each frame is generated independently. By extracting HR using a shorter extraction interval, these inconsistencies can be captured with higher precision, consequently facilitating the distinction between real and fake videos. As a result, our experiment demonstrates that the difference between real and fake videos is most apparent when using *0.1 seconds* in rPPG-based deepfake detection, as HR extracted using *0.1 seconds* can provide abundant temporal information between frames.

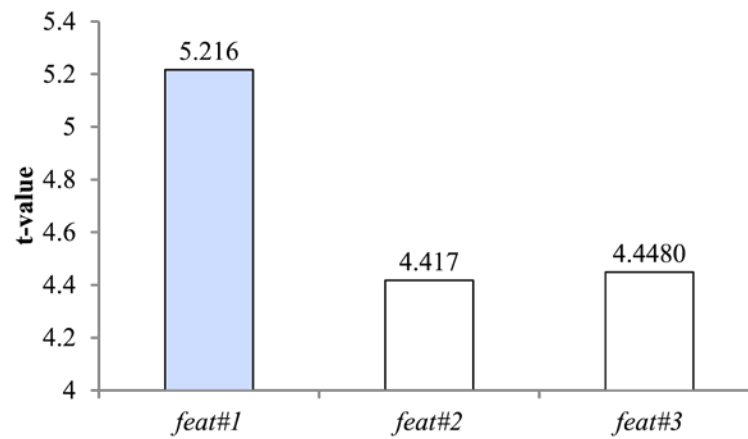
4.1.3 rPPG Extraction Method

We evaluated the differences between real and fake videos in terms of the extracted HR for each rPPG extraction method (i.e., *CHROM*, *GREEN*, *ICA*, *LGI*, *PBV*, and *PCA*). In the independent sample t-test ($p < 0.05$), *GREEN* offered the largest *t*-value of 5.216, as shown in **Table 6** and **Fig. 5**. These results demonstrate that, in rPPG-based deepfake detection, analysis using *GREEN* can more clearly distinguish between real and fake videos compared to analysis using other methods.

Table 7. The detailed results of the t-test on HR values between real and fake videos using different feature engineering methods

Feature Engineering Method	<i>t</i> -value	<i>df</i> [†]	<i>p</i>
<i>feat#1</i>	5.216	125.958	5.53E-07*
<i>feat#2</i>	4.417	149.048	1.82E-05*
<i>feat#3</i>	4.448	150.254	1.6E-05*

* $p < 0.05$, df [†]: degree of freedom

**Fig. 6.** The t-test results on HR values between real and fake videos using different feature engineering methods

The experimental results indicate that *GREEN* had the largest *t*-value (=5.216) because it focuses on the G-channel for extracting plethysmographic signals. The G-channel exhibits a higher absorption of (oxy-) hemoglobin compared to the R-channel and penetrates deeper into the skin compared to the B-channel, providing abundant plethysmographic signal information required for HR extraction [7]. The RGB fusion during the deepfake generation inevitably introduces distortions (e.g., irregularities) to the plethysmographic signal information in the G-channel, consequently facilitating the difference between real and fake videos. As a result, our experiment demonstrates that the difference between real and fake videos is most apparent when using *GREEN* in rPPG-based deepfake detection, as *GREEN* can capture the distorted plethysmographic signal information.

4.1.4 Feature Engineering Method

We evaluated the differences between real and fake videos in terms of the extracted HR for each feature engineering method (i.e., *feat#1*, *feat#2*, and *feat#3*). In the independent sample t-test ($p < 0.05$), *feat#1* offered the largest *t*-value of 5.216, as shown in Table 7 and Fig. 6. These results demonstrate that, in rPPG-based deepfake detection, analysis using *feat#1* can more clearly distinguish between real and fake videos compared to analysis using other methods.

The experimental results indicate that *feat#1* had the largest *t*-value (=5.216) because it focuses on the overall variability of HR [57]. Since the HR patterns in real videos can be disrupted overall during deepfake generation [58], it is crucial to analyze the overall HR

patterns of videos for deepfake detection. However, *feat#2* and *feat#3* primarily analyze the short-term variability of HR by focusing on the differences in adjacent HR. In contrast, *feat#1* measures the overall variability of HR, allowing it to detect non-stationarity in HR patterns that may occur during deepfake generation, consequently facilitating the distinction between real and fake videos. As a result, our experiment demonstrates that the difference between real and fake videos is most apparent when using *feat#1* in rPPG-based deepfake detection, as its extraction can provide non-stationarity of HR patterns in fake videos.

4.2 Key Factor Combination Analysis

In this section, this paper examined a set of key factor combinations and identified the optimal one that maximizes the performance of rPPG-based deepfake detection.

4.2.1 Experimental Setup

Based on the results in section 4.1, we have identified that the following variables maximize the performance of rPPG-based deepfake detection for each key factor: facial region (=R5); extraction interval (=0.1 seconds); rPPG extraction method (=GREEN); and feature engineering method (=feat#1). To confirm whether their combination also offers the most optimal deepfake detection performance, we constructed a total of 540 combinations by considering all the variables of the key factors and measured the difference in HR values between real and fake videos for each combination.

Table 8. Top 20 results of t-test on HR values between real and fake videos for all combinations

No.	Key Factor Combination				t-test for Equality Means		
	Factor#1	Factor#2	Factor#3	Factor#4	t-value	df [†]	p
C#1	R5	0.1 seconds	GREEN	feat#1	5.216	125.958	5.53E-07
C#2	R5	1 second	GREEN	feat#2	5.064	117.748	1.11E-06
C#3	R5	0.5 seconds	GREEN	feat#1	5.006	127.617	1.44E-06
C#4	R5	1 second	GREEN	feat#3	4.978	129.125	1.63E-06
C#5	R5	1 second	GREEN	feat#1	4.771	130.934	4.06E-06
C#6	R5	0.5 seconds	GREEN	feat#3	4.618	136.542	7.85E-06
C#7	R5	0.5 seconds	GREEN	feat#2	4.594	131.348	8.68E-06
C#8	R5	0.1 seconds	GREEN	feat#3	4.448	150.254	1.60E-05
C#9	R5	0.1 seconds	GREEN	feat#2	4.417	149.048	1.82E-05
C#10	R6	0.1 seconds	GREEN	feat#2	4.336	119.015	2.54E-05
C#11	R6	0.1 seconds	GREEN	feat#3	4.26	122.337	3.45E-05
C#12	Avg	0.5 seconds	LGI	feat#1	4.117	161.138	6.09E-05
C#13	R5	0.5 seconds	ICA	feat#3	4.067	157.805	7.42E-05
C#14	Avg	0.5 seconds	CHROM	feat#1	3.976	133.302	1.00E-04
C#15	R5	0.5 seconds	ICA	feat#2	3.935	156.174	1.00E-04
C#16	Avg	0.1 seconds	CHROM	feat#1	3.847	144.27	2.00E-04
C#17	Avg	0.1 seconds	LGI	feat#1	3.842	152.572	2.00E-04
C#18	R4	0.1 seconds	GREEN	feat#1	3.804	130.701	2.00E-04
C#19	Avg	1 second	PBV	feat#3	3.793	122.986	2.00E-04
C#20	R5	1 second	PCA	feat#3	3.766	137.626	2.00E-04

[†]p < 0.05, df[†]: degree of freedom

4.2.2 Experimental Result

Table 8 displays the top twenty combinations with the largest measured t -value. Among these combinations, *C#1*, which consists of the optimal variables identified in this paper offered the largest t -value of 5.216. This result demonstrates that a combination of the optimal variables identified in this paper can also maximize the performance of distinguishing fake videos from real ones in rPPG-based deepfake detection.

Furthermore, this result confirmed that a significant proportion of the top twenty combinations consisted of the optimal variables identified in this paper, indicating that they are crucial for maximizing the difference between real and fake. For example, *C#3*, *C#5*, *C#8*, *C#9*, and *C#18* had the same variables as *C#1* for three key factors (i.e., *C#3* and *C#5*: *R5*, *GREEN*, and *feat#1*, *C#8* and *C#9*: *R5*, *0.1 seconds*, and *GREEN*, and *C#18*: *0.1 seconds*, *GREEN*, and *feat#1*). Similarly, *C#2*, *C#4*, *C#6*, *C#7*, *C#10*, *C#11*, *C#16*, and *C#17* had the same variables as *C#1* for two key factors (i.e., *C#2*, *C#4*, *C#6*, and *C#7*: *R5* and *GREEN*, *C#10* and *C#11*: *0.1 seconds* and *GREEN*, and *C#16* and *C#17*: *0.1 seconds* and *feat#1*). These results demonstrate that the variables identified in this study have a significant impact on rPPG-based deepfake detection performance.

5. Discussion

This study systematically assessed each key factor involved in rPPG-based deepfake detection. To ensure accurate evaluations, we eliminated any external factors that could potentially affect the results. One crucial step in the process of detecting deepfakes based on rPPG is accurately detecting the facial area within a video [59]. However, detecting the facial area can pose a challenge in specific scenarios such as low lighting or partially visible faces. Given that the detection of the facial area is out of our scope, we refined our dataset to ensure effective experiments. Specifically, we filtered our dataset by manually selecting videos that feature a single subject with clearly recognizable eyes, nose, and mouth, as specified in section 3. Additionally, we excluded videos with partially occluded facial areas. This process was essential to ensure that our dataset only included suitable videos for accurate HR value extraction using rPPG. Moreover, the presence of facial hair can limit the area of exposed skin, potentially affecting rPPG-based deepfake detection [60]. Since the extraction of HR from non-skin areas was also out of our scope, we excluded videos containing any facial hair regions that were entirely covered among the nine defined regions (i.e., *R1–R9*). Consequently, our dataset exclusively comprised videos without any completely covered facial hair regions.

This study did not consider the deepfake generation method as the key factor. Experiments were conducted without explicit differentiation between deepfake generation methods, with the aim of identifying universally optimal variables applicable across different methods. Consequently, our dataset was constructed to achieve a balanced ratio between deepfake generation methods (i.e., Deepfakes [11]: 17, Face2Face [61]: 17, FaceSwap [62]: 16, FaceShifter [63]: 16, and NeuralTextures [64]: 16).

This study focuses on identifying and assessing key factors in rPPG-based deepfake detection, rather than introducing new mechanisms. Given this specific focus, there are limited existing approaches available for direct comparison. Furthermore, efforts to apply these key factors to existing methods for rPPG-based deepfake detection were hindered by a scarcity of studies with accessible source code, rendering direct application infeasible. Consequently, in this study, we conducted a comparative analysis based on combinations of these key factors.

6. Conclusion

To improve the effectiveness of rPPG-based deepfake detection, it is important to identify the key factors and analyze their impacts. In this study, we defined four key factors that affect rPPG-based deepfake detection performance: facial region, extraction interval, rPPG extraction method, and feature engineering method. We conducted evaluations for the variables of each factor to comprehend their impact on the performance of rPPG-based deepfake detection. Our experimental results revealed that *R5* (facial region), *0.1 seconds* (extraction interval), *GREEN* (rPPG extraction method), and *feat#1* (feature engineering method) exhibited significant differences in performance between real and fake videos, respectively. Moreover, after analyzing 540 combinations of these variables, we identified that a combination of those optimal variables also offered the most significant distinction between real and fake videos. With these results, this paper successfully confirmed the substantial impact of the variables within each key factor and their combinations on rPPG-based deepfake detection. Our experimental findings enable enhancements in the performance of existing rPPG-based deepfake detection methods by adjusting and applying each identified key factor. Furthermore, the incorporation of the optimal combination of variables can lead to the development of a new rPPG-based deepfake detection framework, providing advanced performance in deepfake detection.

In future work, we plan to design mechanisms aimed at minimizing the influence of external factors, such as lighting conditions, face angles, and facial hair, in rPPG-based deepfake detection. Moreover, we intend to broaden our experiments by incorporating diverse testing environments. Additionally, we will build an rPPG-based deepfake detection framework to which our key factors can be directly applied. Furthermore, we intend to enhance the framework by integrating advanced optimization techniques and supporting newly updated deepfake generation mechanisms. Moreover, we will apply this framework to various types of multimedia applications and assess its usability.

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